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Biological Ageing Pace as a Predictor of Alzheimer's Conversion in MCI: A Statistical and Machine Learning Analysis Using DunedinPACNI and Cognitive Reserve

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Abstract

The regulatory approval of anti-amyloid immunotherapies for early Alzheimer's disease (AD) has made accurate biological risk stratification in the mild cognitive impairment (MCI) stage a clinical priority. Yet despite progress in identifying genetic and pathological predictors of conversion, individual-level prognostic accuracy remains insufficient for clinical deployment, implicating unexplored vulnerability and resilience factors in the heterogeneity of disease trajectories. This dissertation investigates two such constructs: DunedinPACNI (hereafter PACNI), a structural MRI-derived index of multi-system biological ageing pace, and cognitive reserve, operationalised as years of formal education.

Data were drawn from the Alzheimer's Disease Neuroimaging Initiative (ADNI). After application of pre-specified inclusion criteria, the analytic cohort comprised 614 baseline MCI participants (199 converters, 32.4%; 415 stable, 67.6%). Three nested logistic regression models were estimated to quantify the additive and interactive effects of DunedinPACNI and education on three-year conversion, adjusting for age, sex, and APOE $\epsilon 4$ carrier status. Model diagnostics confirmed acceptable multicollinearity (all VIFs < 1.3) and good calibration (Hosmer–Lemeshow $\chi^2(8) = 10.24$, $p = .248$). A supplementary machine learning analysis employing five classifiers under nested five-fold cross-validation evaluated predictive performance across a seven-feature set.

DunedinPACNI was a statistically significant predictor of three-year conversion (OR = 1.25, 95% CI: 1.03–1.51, $p = .022$), with each standard deviation increase in biological ageing pace associated with 25% greater odds of progression. APOE $\epsilon 4$ carrier status was the dominant predictor across all models (OR = 3.13, 95% CI: 2.18–4.54, $p < .001$). Education neither independently predicted conversion (OR = 0.91, $p = .274$) nor moderated the DunedinPACNI–conversion relationship (interaction OR = 1.07, $p = .441$), findings most plausibly attributable to severe restriction of range in educational attainment within this highly educated research cohort (M = 16.0 years, SD = 2.8). DunedinPACNI's predictive contribution was substantially attenuated and no longer statistically significant following adjustment for baseline cognitive performance (OR = 0.98, $p = .882$), indicating that its primary-model association partly reflects contemporaneous disease severity. The primary logistic regression models achieved AUC values of 0.668–0.673;

a seven-feature machine learning model incorporating baseline cognition achieved AUC 0.849–0.870, with the performance gain attributable overwhelmingly to the cognitive composite.

These findings position DunedinPACNI as a biologically meaningful, non-invasive marker of vulnerability in MCI populations that provides prognostic information independent of established genetic risk, whilst acknowledging that its incremental contribution beyond baseline cognitive assessment is modest. They simultaneously identify the conditions under which cognitive reserve testing will fail, including restricted variance in proxy measures and floor effects in compensatory capacity after MCI onset, and argue that future work must employ composite reserve indices and educationally diverse cohorts to resolve the moderation hypothesis definitively. Keywords: Alzheimer’s disease, mild cognitive impairment, DunedinPACNI, biological ageing, cognitive reserve, ADNI, machine learning, MCI-to-AD conversion prediction.

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الْحَمْدُ لِلَّهِ رَبِّ الْعَالَمِينَ

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1 Introduction

1.1 The Clinical Imperative: From Burden to Stratification

Dementia is among the most consequential public health challenges of the twenty-first century, and Alzheimer’s disease (AD) is its dominant form. Approximately 57 million individuals were living with dementia in 2021, a figure projected to exceed 152 million by 2050, driven by population ageing disproportionately concentrated in low- and middle-income countries (World Health Organization, 2021). Alzheimer’s disease accounts for 60–70% of all cases and imposes an escalating burden of morbidity, caregiver dependency, and economic cost: global dementia-related expenditure exceeded US\$1.3 trillion in 2019, with approximately half attributable to unpaid informal care (World Health Organization, 2021). Dementia-related mortality has risen sharply, from approximately 0.56 million deaths in 1990 to 1.62 million in 2019, and the condition now ranks among the leading causes of death worldwide.

For decades, Alzheimer’s disease lacked any disease-modifying treatment. That landscape changed materially in 2023 with the regulatory approvals of lecanemab (Leqembi) and donanemab (Kisunla) for early symptomatic AD in the United States, and subsequently in the United Kingdom, following Phase III trials demonstrating statistically significant slowing of clinical decline (van Dyck et al., 2023; Sims et al., 2023). These approvals mark a genuine therapeutic milestone, but they simultaneously create a new clinical problem: anti-amyloid immunotherapies carry substantial risks, most notably amyloid-related imaging abnormalities (ARIA) including cerebral oedema and microhaemorrhages, with treatment-related deaths reported in trials of both agents. Their benefits, a slowing of decline measured in months rather than years, are modest. The result is that these therapies can only be ethically and cost-effectively deployed in patients whose biological risk profile justifies the exposure. Accurate early stratification — identifying, at the MCI stage, who will progress to AD within a clinically meaningful window — is therefore not merely a scientific aspiration but a prerequisite for the responsible use of these treatments.

This dissertation is motivated by that clinical problem. If progression from MCI to AD is heterogeneous — and it is, with only approximately 10–15% of MCI pa-

tients progressing annually in clinical settings (Mitchell & Shiri-Feshki, 2009), and 32.4% in the present cohort over three years, then understanding the sources of that heterogeneity is prerequisite to stratification. The present work addresses two theoretically grounded candidate sources: accelerated biological ageing pace, indexed by DunedinPACNI, and cognitive reserve, indexed by educational attainment.

1.2 Mild Cognitive Impairment as a Risk State

Mild cognitive impairment is a clinically recognised transitional state between normal cognitive ageing and dementia, characterised by objective cognitive decline that does not yet substantially impair functional independence (Petersen et al., 2018). MCI is profoundly heterogeneous: a substantial proportion of affected individuals remain cognitively stable over extended follow-up periods, and a non-trivial minority revert to normal cognitive functioning, whilst others progress to AD at rates that vary with subtype, biomarker burden, and individual characteristics. Traditional approaches to stratification have relied on established biomarkers: hippocampal atrophy on structural MRI, amyloid positivity on PET imaging, CSF amyloid- β and tau concentrations, and APOE $\epsilon 4$ genotype (Jack et al., 2018). Each predicts conversion, and multimodal approaches improve prognostic performance. Yet individual-level prediction from single or even combined biomarkers remains imperfect, suggesting that progression risk is not fully determined by pathological burden.

1.3 Biological Vulnerability and Cognitive Resilience: Complementary Frameworks

The first construct, biological ageing pace, captures inter-individual variation in the rate of physiological deterioration across organ systems beyond what chronological age predicts. Faster biological ageing is consistently associated with poorer health outcomes (Belsky et al., 2022), and convergent evidence links accelerated systemic ageing to compromised brain health through vascular, neuroimmune, and metabolic pathways (Heneka et al., 2015; Sweeney et al., 2018; Toth et al., 2017). The second construct, cognitive reserve, captures inter-individual resilience: the

capacity to sustain cognitive function in the presence of accumulating neuropathology. Formalised by Stern (2002, 2009), the reserve framework explains the frequently observed dissociation between pathological burden and clinical severity. These constructs are theoretically complementary: biological ageing pace may determine how rapidly pathological burden accumulates, whilst cognitive reserve may determine how much pathological burden must accumulate before clinical expression occurs.

1.4 DunedinPACNI: A Neuroimaging Index of Biological Ageing Pace

Whitman et al. (2025) introduced DunedinPACNI (Dunedin Pace of Aging Calculated from NeuroImaging; hereafter PACNI), a structural MRI-derived biomarker designed to estimate longitudinal biological ageing from a single brain scan. The measure was developed using data from the Dunedin Study, where an elastic net model was trained on 315 MRI-derived structural features to predict Dunedin Study’s longitudinal Pace of Aging measure (Belsky et al., 2022), itself derived from repeated assessment of 19 biomarkers spanning cardiovascular, metabolic, renal, immune, dental, and pulmonary functioning across approximately two decades. Unlike conventional brain-age models, which estimate chronological age from cross-sectional MRI data, DunedinPACNI was developed to estimate the rate of biological ageing itself. Initial validation in ADNI ($n = 701$) demonstrated that participants with MCI who exhibited faster DunedinPACNI at baseline were significantly more likely to convert to dementia ($HR = 1.44$, 95% CI: 1.26–1.65, $p < .001$), an association that remained largely unchanged after adjustment for APOE $\epsilon 4$ allele status ($HR = 1.42$, 95% CI: 1.23–1.62, $p < .001$), suggesting that DunedinPACNI captures prognostic information beyond established genetic risk factors.

1.5 Research Rationale and Thesis Structure

Despite its promising initial validation, Whitman et al. (2025) did not examine cognitive reserve as a modifying construct. If conversion risk is jointly determined by biological vulnerability and cognitive resilience, as the diathesis-stress framework predicts, then models omitting their interaction are theoretically incomplete.

This dissertation therefore pursues four objectives: to confirm whether DunedinPACNI independently predicts three-year MCI-to-AD conversion in ADNI; to examine whether educational attainment independently predicts conversion; to formally test whether education moderates the DunedinPACNI–conversion relationship; and to benchmark predictive performance using five machine learning classifiers with nested cross-validation. Chapter 2 specifies the research gap, aims, and statistical framework; Chapter 3 reviews the relevant literature; Chapter 4 describes the methods; Chapter 5 presents results; Chapters 6 and 7 discuss and conclude.

2 Research Gap, Aims, and Objectives

2.1 The Research Gap

The literature establishes three relevant partial findings. First, DunedinPACNI predicts MCI conversion independently of APOE $\epsilon 4$ status (Whitman et al., 2025). Second, cognitive reserve, indexed by educational attainment, modifies AD risk and the pathology–cognition relationship in several cohorts (Valenzuela & Sachdev, 2006; Roe et al., 2008). Third, the diathesis-stress model predicts that the magnitude of a biological risk factor’s effect on outcome should be attenuated in individuals with greater reserve capacity (Monroe & Simons, 1991). What has not been tested is whether reserve moderates the DunedinPACNI–conversion relationship specifically. Whitman et al. (2025) did not examine reserve-related variables as moderators, did not construct composite CR measures, and did not test whether PACNI’s predictive value is conditional on compensatory capacity. The present dissertation closes this gap.

A secondary gap concerns the machine learning literature on MCI conversion prediction. The majority of published models treat biomarkers as fixed predictors and report performance figures derived from single-split or non-nested cross-validation frameworks, which are known to produce optimistically biased AUC estimates (Cawley & Talbot, 2010). The present work employs a methodologically rigorous nested five-fold framework and includes DunedinPACNI as a feature alongside established clinical and genetic predictors, providing an unbiased estimate of the biomarker’s contribution to discriminative performance.

2.2 Primary Research Aim

To determine whether DunedinPACNI independently predicts three-year MCI-to-AD conversion in the ADNI cohort, and whether educational attainment moderates this relationship as a cognitive reserve proxy; and whether a machine learning approach employing biological ageing pace achieves clinically meaningful discriminative performance.

2.3 Central Hypothesis

The primary hypothesis was that higher DunedinPACNI would be associated with increased conversion risk (positive main effect) and that this association would be attenuated in individuals with greater educational attainment (negative interaction coefficient), consistent with an additive buffering model of cognitive reserve. Secondary hypotheses specified that education would show a negative independent association with conversion risk, and that DunedinPACNI would contribute incrementally to predictive accuracy within a machine learning framework.

2.4 Statistical Framework

Three nested logistic regression models were estimated sequentially. The full Model 3 specification is:

$$\begin{aligned} \text{logit}(P_i) = & \beta_0 + \beta_1 \text{PACNI}_i + \beta_2 \text{Edu}_i \\ & + \beta_3 (\text{PACNI}_i \times \text{Edu}_i) + \beta_4 \text{Age}_i \\ & + \beta_5 \text{Sex}_i + \beta_6 \text{APOE}\epsilon 4_i \end{aligned} \quad (1)$$

where P_i denotes the probability of three-year MCI-to-AD conversion for participant i . In the interaction model, β_1 represents the effect of DunedinPACNI when educational attainment is held at its mean value ($z = 0$), β_2 represents the effect of educational attainment when DunedinPACNI is held at its mean value, and β_3 represents the interaction effect testing whether the association between DunedinPACNI and conversion risk varies as a function of educational attainment. Age, sex, and APOE $\epsilon 4$ carrier status were included as covariates on the basis of their established associations with AD risk. All continuous predictors were standardised

to z-scores ($M = 0$, $SD = 1$) prior to analysis, facilitating comparison of effect sizes and reducing non-essential multicollinearity between lower-order terms and their interaction (Kraemer & Blasey, 2004).

2.5 Research Objectives

1. **Objective 1** Validate DunedinPACNI as a significant independent predictor of three-year MCI-to-AD conversion after adjustment for age, sex, and APOE $\epsilon 4$ carrier status.
2. **Objective 2** Examine whether years of formal education independently predicts three-year conversion and improves model fit beyond DunedinPACNI and covariates.
3. **Objective 3** Formally test whether educational attainment moderates the association between DunedinPACNI and conversion risk.
4. **Objective 4** Benchmark predictive performance using five machine learning classifiers with nested five-fold cross-validation, assessing the incremental contribution of DunedinPACNI.
5. **Objective 5** Contextualise primary findings through supplementary analyses incorporating baseline cognitive performance and CSF biomarker adjustment.

3 Literature Review

3.1 Overview

This chapter reviews the theoretical and empirical foundations of the present research across four interrelated domains: the conceptualisation, operationalisation, and epidemiological evidence for cognitive reserve; the measurement of biological ageing pace and its relationship to brain health; computational approaches to MCI progression prediction; and the integration of vulnerability and resilience constructs in dementia risk modelling.

3.2 Cognitive Reserve in Ageing and Dementia

3.2.1 Conceptual Origins and Theoretical Framework

The cognitive reserve framework is grounded in a clinicopathological paradox that has resisted simple explanation for over four decades. Post-mortem studies beginning with Katzman et al. (1988) and Blessed et al. (1968) established a consistent and disturbing dissociation: individuals meeting full neuropathological criteria for Alzheimer’s disease at autopsy had, in many cases, shown minimal cognitive impairment during life. This pattern cannot be explained by measurement error, selection bias, or post-mortem artefact alone, because it has been replicated across independent cohorts using standardised neuropathological criteria and prospective cognitive assessments. One influential inference is that the relationship between pathological burden and clinical expression is moderated by individual difference factors, implying a moderation account rather than a purely main effect account.

Stern (2002, 2009) formalised this into the cognitive reserve construct, distinguishing it carefully from brain reserve (the passive threshold model, in which larger brains tolerate more damage before a clinical threshold is crossed) by emphasising active functional compensation. High-reserve individuals recruit alternative neural networks or deploy more efficient processing strategies when primary circuits are compromised, thereby maintaining clinical function at pathological burden levels that would impair low-reserve individuals. This distinction has important implications for operationalisation: brain reserve is plausibly indexed by head circumference or synaptic count, whereas cognitive reserve requires indices of functional capacity or lifetime cognitive engagement. The near-universal reliance on years of formal education as a reserve proxy in epidemiological studies, including the present work, is therefore a pragmatic concession rather than a theoretically optimal solution. Education is a distal, time-invariant proxy that conflates the age at which schooling was received, its quality, and the degree to which formal schooling translates into lasting neural enrichment. The consensus view, as formalised by Stern et al. (2020), is that education should be treated as one indicator within a composite reserve index rather than as a sufficient operationalisation — a limitation that has direct consequences for the statistical power available in the present study.

3.2.2 Neurobiological Mechanisms

The proposed biological mechanisms of cognitive reserve can be organised around three non-mutually-exclusive accounts: neural efficiency, neural compensation, and structural enrichment. Barulli and Stern (2013) proposed that high-reserve individuals show less neural resource expenditure for equivalent task performance under conditions of intact brain function (efficiency), but can recruit additional or alternative networks when primary circuits are impaired (compensation). Franzmeier et al. (2020) provided in vivo support for the compensation account: degree of functional network connectivity predicted the rate of tau accumulation across Alzheimer's disease stages, suggesting that network architecture actively shapes disease trajectory rather than merely reflecting it. However, these studies share a critical methodological limitation: they are conducted in samples where reserve level has already been developed by the time of measurement, making it impossible to determine whether observed neural differences are a consequence of reserve capacity accumulated over the lifespan, or a concurrent feature of the neurological state being studied.

The structural enrichment hypothesis proposes that cognitively stimulating experiences increase neural complexity and raise the threshold at which pathology produces clinical impairment. Evidence from post-mortem studies and experience-dependent plasticity research supports this account, but much of the evidence derives from highly selected groups and cross-sectional designs, limiting generalisability and causal inference. Longitudinal studies examining neural change as a function of cognitive engagement remain relatively limited and have not consistently confirmed the causal direction implied by the reserve hypothesis.

3.2.3 Epidemiological Evidence and MCI Progression

The epidemiological evidence for a protective effect of cognitive reserve is extensive. Meta-analytic evidence indicates that low educational attainment is associated with elevated dementia risk; Valenzuela's and Sachdev's (2006) meta-analysis of 22 prospective studies found that individuals with high levels of mental activity exhibited approximately 46% lower dementia risk relative to those with low mental activity. Crucially, several studies suggest that reserve proxies modify the relationship between pathology and cognition: among individuals with equivalent

amyloid burden, those with greater educational attainment or occupational complexity often maintain better cognitive performance, findings consistent with a buffering rather than a purely preventive mechanism.

3.2.4 Biological Ageing Measures and Brain Health

The brain does not age in isolation from systemic physiology. Multiple pathways link accelerated systemic ageing to compromised brain health: vascular ageing impairs the brain's metabolic supply (Toth et al., 2017); chronic low-grade inflammation promotes microglial activation and may accelerate tau phosphorylation and amyloid accumulation (Heneka et al., 2015); and blood-brain barrier degradation permits infiltration of peripheral immune factors that can exacerbate neuroinflammation (Sweeney et al., 2018). These mechanisms provide a biological rationale for examining DunedinPACNI as a potential predictor of AD progression.

3.2.5 Epigenetic Clocks and Biological Ageing Measurement

The most influential approach to quantifying biological ageing has been the development of epigenetic clocks based on DNA methylation patterns. Horvath (2013) demonstrated that a weighted sum of CpG methylation values could predict chronological age with a median absolute error of approximately 3.6 years across diverse tissues. Second-generation clocks including PhenoAge (Levine et al., 2018) and GrimAge (Lu et al., 2019) were trained to predict physiological ageing markers and mortality rather than chronological age, exhibiting stronger associations with health outcomes. The DunedinPACE measure (Belsky et al., 2022) extended this framework by quantifying the rate of biological ageing directly, estimating not how old someone appears biologically but how fast they are ageing. DunedinPACNI extends the pace-of-ageing framework to the neuroimaging domain by deriving an estimate of biological ageing pace from structural MRI data.

3.2.6 Critical Appraisal of DunedinPACNI as a Biomarker of Biological Ageing

DunedinPACNI offers several advantages over existing biological ageing measures that deserve explicit acknowledgement. Unlike blood-based epigenetic clocks,

which require biofluid collection and specialised assay infrastructure, DunedinPACNI is derivable from standard structural MRI already acquired as part of routine clinical neuroimaging workflows. Unlike single-organ neurodegeneration markers (hippocampal volume, cortical thickness, white matter hyperintensity burden), it was trained to index multi-system biological ageing pace rather than brain-specific pathology, providing a theoretically distinct signal. Its cross-system sensitivity is evidenced by the breadth of its UK Biobank associations: mortality, multimorbidity, frailty, grip strength, and hippocampal atrophy are not obviously reducible to a single biological mechanism, suggesting that DunedinPACNI may capture aspects of organismic ageing rather than a specific pathological signature.

However, several validity concerns require careful consideration. The most substantive is the conflation of ageing pace with disease severity in a late-life clinical population. DunedinPACNI was trained on a general-population cohort aged 26–45 at measurement, where biological ageing pace and neurodegenerative pathology are statistically independent: the sample contained no participants with MCI or dementia, and the training signal reflected genuine variation in multi-system physiological deterioration rates. In a late-life MCI sample, the interpretive context is substantially different: structural MRI features that index DunedinPACNI — including cortical thinning, hippocampal atrophy, and white matter degradation — are also direct indices of Alzheimer’s neuropathology. The elastic net model trained on healthy adults may therefore, when applied to an MCI sample, partially recapitulate a disease severity index. This confound cannot be resolved with cross-sectional data; it requires either (a) measuring DunedinPACNI at multiple time points before MCI onset and examining whether the rate of change in PACNI scores, independent of their level, predicts subsequent conversion, or (b) testing PACNI’s predictive validity in amyloid-negative MCI participants, for whom neurodegeneration-related structural changes should be minimal.

A second concern involves standardisation and pipeline dependence. DunedinPACNI’s elastic net coefficients were estimated using FreeSurfer output from the Dunedin Study, which used a specific scanner, protocol, and preprocessing pipeline. Applying the model to ADNI — which spans four study phases, multiple scanner manufacturers, and both 1.5T and 3T field strengths — introduces systematic measurement variation that the original model was not designed to accommodate. Whitman et al. (2025) addressed this partially by harmonising ADNI

data prior to PACNI computation, but the degree to which harmonisation fully eliminates protocol-related variance is unknown. Scanner-related confounds could either attenuate true predictive signal (adding noise) or create spurious signal (if scanner-related structural differences correlate with diagnostic status). Third, the generalisability of DunedinPACNI beyond the populations in which it has been validated — predominantly European-ancestry New Zealand and UK cohorts — remains to be established. Structural brain architecture varies across ancestral populations due to both genetic and environmental factors, and elastic net coefficients trained in one population may not generalise to another without recalibration. A fourth and more fundamental concern is causal interpretation. Even if DunedinPACNI validly measures multi-system biological ageing pace, the relationship between ageing pace and AD conversion risk may be confounded by shared genetic architecture. APOE $\epsilon 4$ not only increases AD risk directly but also accelerates systemic biological ageing (via lipid metabolism and inflammatory pathway dysregulation), meaning that partial APOE $\epsilon 4$ independence of the PACNI effect, as demonstrated in the present study and by Whitman et al. (2025), does not fully exclude genetic confounding. Full exclusion would require testing PACNI's predictive validity after adjusting for the full genomic principal components, a step that the present study did not take due to data constraints. These limitations do not undermine the empirical finding of a statistically significant PACNI effect, but they do constrain its interpretive range: the present results support DunedinPACNI as a biomarker of conversion risk, not as a validated causal mechanism in the chain between biological ageing and Alzheimer's disease.

3.2.7 Computational Approaches to MCI Progression Prediction

The machine learning literature on MCI-to-AD conversion prediction has matured considerably over fifteen years but is characterised by optimism bias and methodological inconsistency. Davatzikos et al. (2011) demonstrated that structural MRI features could predict conversion with approximately 75–80% accuracy using SVM classifiers without nested cross-validation. Subsequent multimodal studies combining MRI, PET, CSF, and genetic biomarkers typically yielded AUC values of 0.85–0.92, though many evaluated performance on hold-out test sets that were not completely held out during preprocessing or feature selection, resulting in

optimistic estimates. Battista et al. (2020) systematically reviewed 83 MCI classification studies and identified a systematic relationship between methodological rigour and reported performance. Wen et al. (2020) demonstrated this empirically: correcting for data leakage reduced apparent performance by 8–12 AUC points. These findings contextualise the present study’s conservative approach of nested cross-validation, and its primary logistic regression AUC of 0.668 as the uncontaminated contribution of biological ageing and genetic risk without clinical severity information.

3.2.8 Integrating Vulnerability and Resilience in Dementia Risk Modelling

The integration of biological vulnerability and cognitive resilience into a single predictive framework represents the theoretical frontier of the MCI progression literature. The diathesis-stress model (Monroe & Simons, 1991) predicts a specific interaction: biological ageing pace should have the greatest adverse effect in individuals with low cognitive reserve, who lack the compensatory neural capacity to buffer accumulating neurodegenerative damage, and a reduced or absent effect in high-reserve individuals. The strongest available evidence for this class of interaction comes from studies examining APOE ϵ 4-by-education moderation: Roe et al. (2008) showed that APOE ϵ 4’s adverse effect on cognitive trajectory was significantly attenuated in individuals with high educational attainment, consistent with a reserve-as-buffer model. However, neither that study nor Marden et al. (2017) explicitly tested neuroimaging-derived biological ageing pace as the diathesis, and both were conducted in population-based samples with substantially greater educational variance than the present ADNI cohort. The present study is, to the author’s knowledge, the first to formally test whether a neuroimaging-based biological ageing rate and an educational reserve proxy interact to predict MCI-to-AD conversion — a gap that reflects not a lack of theoretical motivation but the recency of DunedinPACNI’s availability (Whitman et al., 2025).

4 Method

4.1 Data Source

Data were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI), a longitudinal multi-site study established in 2003 to develop and validate clinical, imaging, genetic, and biochemical biomarkers for the early detection and monitoring of Alzheimer’s disease (Petersen et al., 2010; Weiner et al., 2017). Data are publicly available at adni.loni.usc.edu following approval. The merged dataset comprised 8,428 participant visits from 2,177 unique individuals across ADNI-1, ADNI-GO, ADNI-2, and ADNI-3 phases. ADNI has received ethical approval from the institutional review boards of all participating sites, and all participants provided written informed consent prior to enrolment.

4.2 Participant Selection

Inclusion criteria were pre-specified as follows: (1) baseline diagnosis of mild cognitive impairment (Late MCI or Early MCI); (2) availability of a valid structural MRI scan at the baseline visit enabling DunedinPACNI computation; and (3) at least one follow-up visit within the three-year window (Months 24, 30, 36, or 42) permitting determination of conversion status. Of 2,177 unique individuals in the ADNI dataset, 2,083 had available baseline DunedinPACNI scores. Of these, 961 met MCI diagnostic criteria. Following application of the three-year follow-up availability criterion, 347 participants were excluded, yielding a final analytic cohort of 614 participants.

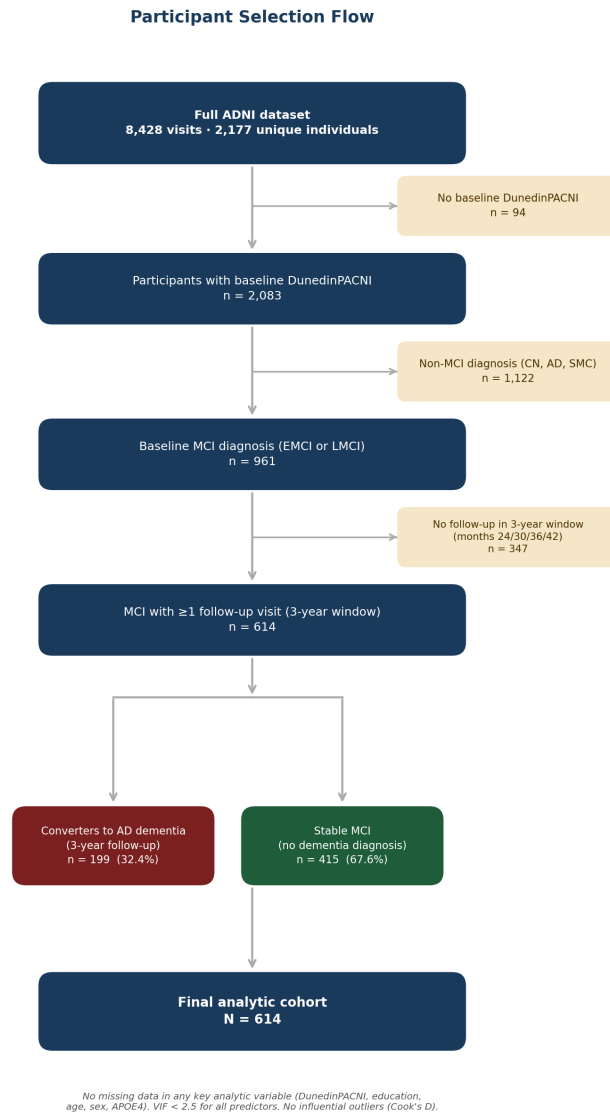


Figure 1: Participant selection flow diagram. Numbers represent unique participants at each stage. The final analytic cohort (N = 614) comprises all participants with a baseline MCI diagnosis, an available DunedinPACNI score, and at least one follow-up visit within the three-year window.

4.3 DunedinPACNI

DunedinPACNI scores were computed using the algorithm published by Whiteman et al. (2025), implemented via the open-source R package available at github.com/etw11/DunedinPACNI. The algorithm applies an elastic net regression model trained on 315 MRI-derived structural measures — including regional cortical thickness, subcortical volumes, and white matter characteristics processed through FreeSurfer (version 6.0) — to estimate individual pace of multi-system biological ageing from a single cross-sectional scan. Only baseline visit scores were retained. DunedinPACNI is scaled such that 1.0 represents normative ageing pace; values above 1.0 indicate faster biological ageing relative to the training population.

4.4 Outcome and Predictors

The primary outcome was binary three-year conversion to Alzheimer’s disease dementia. A participant was classified as a converter if they received a clinical diagnosis of dementia at any follow-up visit within the three-year window. A composite baseline clinical burden score was constructed from four core clinical measures: the Mini-Mental State Examination (MMSE), Clinical Dementia Rating Sum of Boxes (CDRSB), Functional Activities Questionnaire (FAQ), and the Alzheimer’s Disease Assessment Scale-Cognitive Subscale 13-item version (ADAS-13). Years of formal education served as the primary cognitive reserve proxy, in accordance with Stern et al. (2020). All continuous predictors were standardised to z-scores prior to analysis.

4.5 Statistical Analysis

All primary analyses were conducted in R (version 4.3.1). Three logistic regression models were estimated using the `glm()` function with a binomial family and logit link. Variance inflation factors (VIF) were computed using the `car` package to assess multicollinearity (all VIFs < 1.3 in primary models). Goodness-of-fit was assessed using the Hosmer–Lemeshow test ($\chi^2(8) = 10.24$, $p = .248$ for Model 1). With 199 events and up to 7 parameters, the events-per-variable ratio was 33.2, well above the accepted minimum of 10. Model fit was compared using

the Akaike Information Criterion (AIC) and formal likelihood ratio tests (LRT). Discriminative performance was quantified using the area under the receiver operating characteristic curve (AUC), with confidence intervals estimated by DeLong's method.

4.6 Machine Learning Analysis

A complementary machine learning analysis was conducted in Python 3.11 using scikit-learn (v1.3) and XGBoost (v2.0). Five classifiers were evaluated: logistic regression (LR), random forest (RF), support vector machine (SVM), k-nearest neighbours (KNN), and XGBoost (XGB). The feature matrix comprised seven variables: DunedinPACNI, education, age, sex, APOE ϵ 4 carrier status, MCI subtype, and the composite baseline cognition score. A true nested five-fold cross-validation framework was employed to reduce optimism bias in performance estimation by fully separating hyperparameter tuning from performance evaluation, in which a GridSearchCV object with an inner five-fold StratifiedKFold was passed directly to cross_validate with an outer five-fold StratifiedKFold, ensuring hyperparameter selection occurred independently within each outer fold (Cawley & Talbot, 2010). Class imbalance was addressed by applying class_weight=balanced for LR, RF, and SVM; for XGBoost, scale_pos_weight was set to 2.09, derived from the inverse class ratio (415 stable / 199 converters), consistent with the weighting applied to the other classifiers. An incremental feature contribution analysis was conducted to formally quantify the marginal AUC gain attributable to each feature group.

5 Results

5.1 Sample Characteristics

Variable	Converters (n = 199)	Stable MCI (n = 415)	Total (N = 614)
Age, years: mean (SD)	73.5 (7.2)	72.4 (7.6)	72.8 (7.5)
Male sex, n (%)	120 (60.3%)	257 (61.9%)	377 (61.4%)
Education, years: mean (SD)	15.8 (2.8)	16.1 (2.8)	16.0 (2.8)
APOE ϵ 4 carrier, n (%)	136 (68.3%)	178 (42.9%)	314 (51.1%)
Late MCI, n (%)	183 (92.0%)	220 (53.0%)	403 (65.6%)
DunedinPACNI: mean (SD)	0.947 (0.503)	0.845 (0.485)	0.879
Baseline Cognition Score: mean (SD)	0.582 (0.659)	-0.278 (0.559)	0.001 (-1.32-2.74)

Table 1: Weighted cohort mean. DunedinPACNI scored such that 1.0 = normative ageing pace. Baseline Cognition Score is a composite of reversed MMSE, CDRSB, FAQ, and ADAS-13; used in supplementary analyses only.

The cohort comprised 403 participants with Late MCI (65.6%) and 211 with Early MCI (34.4%). The mean age was 72.8 years (SD = 7.5), 61.4% were male, and mean years of education were 16.0 (SD = 2.8). APOE ϵ 4 carrier status was present in 314 participants (51.1%). Over the three-year follow-up, 199 participants (32.4%) converted to Alzheimer's disease dementia, whilst 415 (67.6%) remained stable.

5.2 DunedinPACNI Across the Alzheimer's Disease Continuum

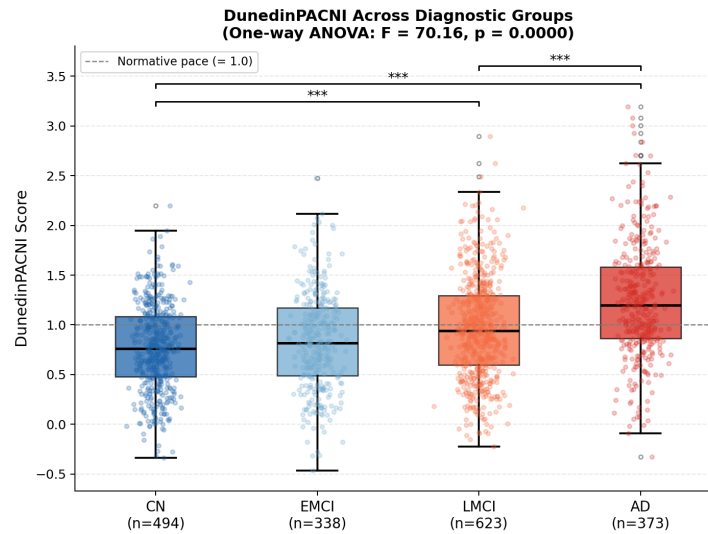


Figure 2: Distribution of DunedinPACNI scores across ADNI diagnostic groups ($N = 1,828$ with available baseline scores). The dashed line at 1.0 denotes normative ageing pace. *** $p < .001$.

To contextualise the primary analytic cohort within the broader disease spectrum, DunedinPACNI scores were examined across the four principal ADNI diagnostic categories in a baseline diagnostic-group sample ($N = 1,828$). Although baseline DunedinPACNI scores were available for 2,083 participants, this analysis was restricted to individuals classified at baseline as cognitively normal (CN), early mild cognitive impairment (EMCI), late mild cognitive impairment (LMCI), or Alzheimer's disease/dementia (AD); participants with other baseline classifications, such as significant memory concern (SMC), were excluded. A one-way ANOVA revealed a highly significant group effect, $F(3, 1824) = 70.16$, $p < .001$. Mean scores increased progressively from CN ($M = 0.768$, $SD = 0.428$) through EMCI ($M = 0.832$, $SD = 0.488$) and LMCI ($M = 0.966$, $SD = 0.509$) to AD ($M = 1.240$, $SD = 0.576$), with Cohen's d for the CN–AD contrast reaching 0.95. Holm-corrected pairwise t -tests confirmed significant differences for all pairs except CN–EMCI ($p_{\text{raw}} = .048$, non-significant after Holm correction), consistent with the R-based Tukey analysis ($p = .277$).

5.3 Primary Logistic Regression Analysis

5.3.1 Model 1: DunedinPACNI as a Predictor of Conversion

DunedinPACNI was a statistically significant predictor of three-year conversion after controlling for age, sex, and APOE $\epsilon 4$ carrier status (OR = 1.25, 95% CI: 1.03–1.51, $p = .022$). Each one standard deviation increase in DunedinPACNI was associated with 25% greater odds of progression from MCI to dementia. APOE $\epsilon 4$ carrier status was the strongest predictor in the model (OR = 3.13, 95% CI: 2.18–4.54, $p < .001$). Age (OR = 1.16, $p = .150$) and sex (OR = 0.84, $p = .349$) were not independently associated with conversion. Model fit was acceptable (AIC = 736.69), and discrimination was modest (AUC = 0.668, 95% CI: 0.622–0.714). Model diagnostics indicated adequate calibration (Hosmer–Lemeshow $\chi^2(8) = 10.24$, $p = .248$) and no evidence of problematic multicollinearity (all VIFs < 1.3).

5.3.2 Model 2: Adding Education as a Cognitive Reserve Proxy

Adding years of formal education to the model did not significantly improve model fit (LRT: $\chi^2(1) = 1.19$, $p = .275$; AIC increased from 736.69 to 737.49). Education was not a statistically significant predictor of three-year conversion (OR = 0.91, 95% CI: 0.76–1.08, $p = .274$), though the direction was negative, consistent with the reserve hypothesis. The DunedinPACNI effect remained essentially unchanged (OR = 1.25, 95% CI: 1.04–1.52, $p = .021$). The DeLong test for Model 1 versus Model 2 confirmed no significant improvement in discrimination ($Z = -0.61$, $p = .539$; AUC M2 = 0.671).

5.3.3 Model 3: DunedinPACNI × Education Interaction

Variable	Model 1 OR (95% CI)	Model 2 OR (95% CI)	Model 3 OR (95% CI)
DunedinPACNI	1.25 (1.03–1.51)*	1.25 (1.04–1.52)*	1.24 (1.03–1.51)*
Education	—	0.91 (0.76–1.08)	0.90 (0.75–1.07)
PACNI × Education	—	—	1.07 (0.90–1.29)
APOE ε4 carrier	3.13 (2.18–4.54)***	3.12 (2.17–4.52)***	3.11 (2.17–4.51)***
Age	1.16 (0.95–1.41)	1.15 (0.94–1.40)	1.14 (0.94–1.39)
Sex (Male)	0.84 (0.58–1.21)	0.87 (0.60–1.26)	0.87 (0.60–1.26)
AIC	736.69	737.49	738.90
AUC (95% CI)	0.668 (0.622–0.714)	0.671 (0.625–0.716)	0.673 (0.628–0.718)
LRT vs prior	—	$\chi^2(1)=1.19, p=.275$	$\chi^2(1)=0.59, p=.441$

Table 2: All continuous predictors standardised to z-scores. * $p < .05$; *** $p < .001$. AUC = area under the ROC curve (DeLong method). Model 1 Hosmer–Lemeshow $\chi^2(8) = 10.24, p = .248$; all VIFs < 1.3 .

The interaction between DunedinPACNI and education was not statistically significant (OR = 1.07, 95% CI: 0.90–1.29, $p = .444$). The interaction term did not improve model fit relative to Model 2 (LRT: $\chi^2(1) = 0.59, p = .441$), and the AIC increased further to 738.90. DunedinPACNI remained a statistically significant predictor of conversion (OR = 1.24, 95% CI: 1.03–1.51, $p = .025$).

5.4 Supplementary Analyses

5.4.1 Supplementary A: Baseline Cognition Adjustment

When the composite baseline cognition score was added to Model 1, the independent predictive contribution of DunedinPACNI was substantially attenuated and no longer reached conventional significance (SA1: OR = 0.98, 95% CI: 0.78–1.23, $p = .882$). The cognition composite itself was a highly significant predictor of conversion (OR = 4.84, 95% CI: 3.71–6.47, $p < .001$), consistent with its role as an index of baseline clinical severity. APOE ε4 remained significant (OR = 2.54, 95% CI: 1.65–3.95, $p < .001$). AUC values for the cognition-adjusted models were

substantially higher (SA1–SA3: AUC = 0.855–0.856) than the primary models, driven by the strong predictive value of the cognitive composite.

5.4.2 Supplementary B: CSF Biomarker Subsample

Analysis	DunedinPACNI OR (95% CI)	p	Key Note
SA1: + Baseline Cognition	0.98 (0.78–1.23)	.882	Cognition OR = 4.84 (3.71–6.47)***
SA2: + Education + Cognition	0.98 (0.78–1.23)	.844	Education non-sig (p = .379)
SA3: Interaction + Cognition	0.98 (0.78–1.23)	.847	Interaction non-sig (p = .866)
SB1: Biomarker subset (n = 339)	1.10 (0.85–1.42)	.464	APOE ϵ 4 OR = 2.53 (1.56–4.16)***
SB2: + CSF biomarkers	1.18 (0.89–1.56)	.252	Tau/p-tau VIF > 25; CSF ORs unreliable

Table 3: SB2 exhibited severe multicollinearity between total tau and p-tau181 ($r = .984$; VIFs = 25.84 and 25.38). *** $p < .001$.

In the biomarker subsample with available CSF data ($n = 339$), DunedinPACNI did not reach statistical significance either before (SB1: OR = 1.10, $p = .464$) or after (SB2: OR = 1.18, $p = .252$) adjustment for CSF biomarkers ($A\beta_{42}$, total tau, and p-tau181). An important caveat applies to the SB2 model: total tau and p-tau181 were severely multicollinear ($r = .984$; VIF = 25.84 and 25.38, respectively), far exceeding conventional thresholds. This multicollinearity renders the individual tau and p-tau coefficients in SB2 statistically unreliable. The overall LRT confirming the joint contribution of the three CSF markers remains valid ($\Delta\text{deviance} = 66.03$, $df = 3$, $p < .001$), but individual tau and p-tau estimates should not be interpreted independently.

5.5 Machine Learning Results

5.5.1 Classifier Performance

Model	AUC-ROC	Accuracy	F1 Score	Sensitivity	Precision
Logistic Regression	0.867 ± 0.024	0.749 ± 0.041	0.677 ± 0.038	0.804 ± 0.058	0.588 ± 0.058
Random Forest	0.861 ± 0.022	0.791 ± 0.021	0.708 ± 0.026	0.779 ± 0.049	0.652 ± 0.043
SVM	0.870 ± 0.024	0.774 ± 0.026	0.695 ± 0.030	0.794 ± 0.049	0.620 ± 0.041
KNN	0.849 ± 0.016	0.779 ± 0.009	0.610 ± 0.041	0.543 ± 0.082	0.713 ± 0.038
XGBoost	0.856 ± 0.028	0.780 ± 0.025	0.703 ± 0.032	0.804 ± 0.067	0.629 ± 0.046

Table 4: DunedinPACNI, education, age, sex, APOE ϵ 4, MCI subtype, baseline cognition (7 features). The substantially higher AUC relative to the primary logistic regression models (AUC = 0.668–0.673) primarily reflects the inclusion of baseline cognitive performance.

All five classifiers demonstrated strong discriminative performance on the seven-feature set under nested five-fold cross-validation. AUC-ROC values ranged from 0.849 ± 0.016 (KNN) to 0.870 ± 0.024 (SVM), substantially exceeding the 0.668 achieved by the primary logistic regression model using four predictors. SVM achieved the highest AUC (0.870 ± 0.024), closely followed by logistic regression (0.867 ± 0.024), suggesting that the relationship between predictors and conversion risk may be captured effectively by relatively simple decision boundaries. Random Forest achieved the highest accuracy (0.791 ± 0.021) and F1 score (0.708 ± 0.026), indicating the strongest balance between sensitivity and precision overall. Logistic Regression and XGBoost jointly achieved the highest sensitivity (0.804 ± 0.058 and 0.804 ± 0.067 , respectively), a particularly important metric for identifying individuals at risk of conversion to Alzheimer’s disease dementia.

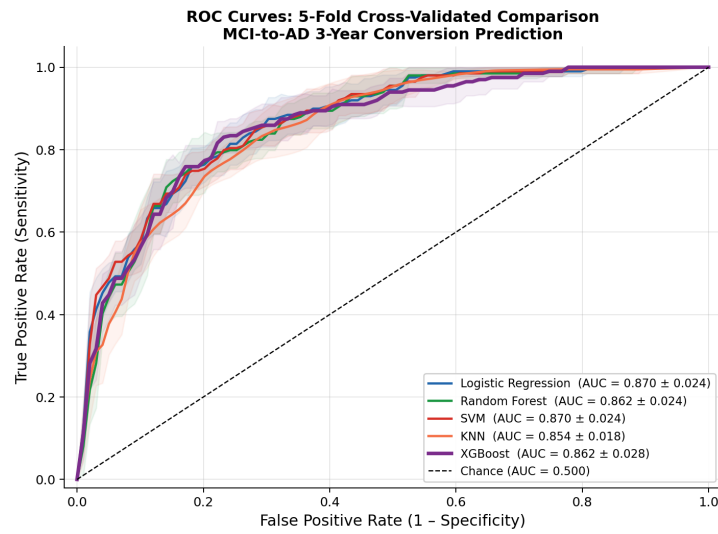


Figure 3: Receiver operating characteristic curves for all five classifiers (5-fold cross-validated). The substantially higher AUC relative to the primary logistic regression models (0.668) reflects the inclusion of baseline cognition in the machine learning feature set.

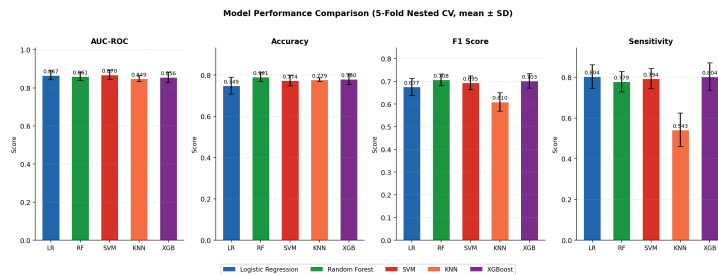


Figure 4: Model performance comparison across five metrics (5-fold CV, mean $\pm$ SD). SVM achieved the highest AUC (0.870); XGBoost achieves the highest F1 and accuracy.

5.5.2 Feature Importance Analysis

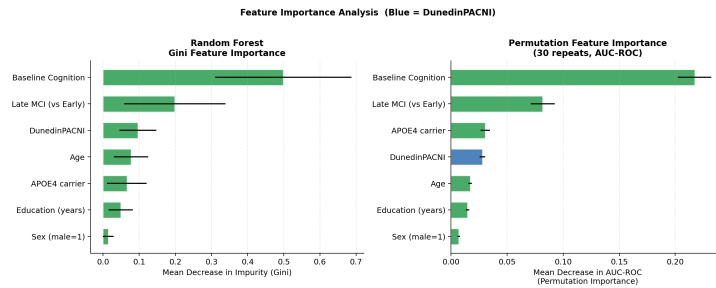


Figure 5: Random Forest Gini impurity importance and permutation feature importance (30 repeats, AUC-ROC) both ranked baseline cognition as the dominant contributor, followed by MCI subtype. DunedinPACNI ranked fourth in permutation importance and third in Gini importance, ahead of age, APOE ε4 status, education, and sex.

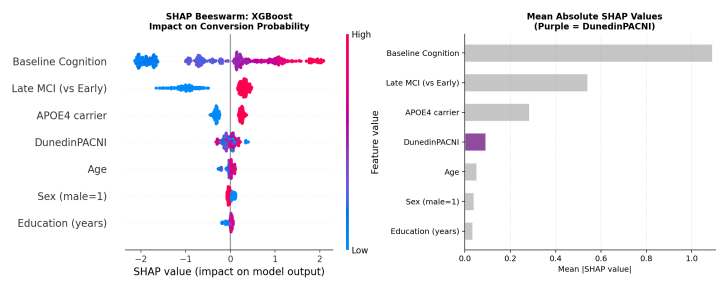


Figure 6: SHAP analysis for XGBoost confirmed that DunedinPACNI made a modest but directionally consistent positive contribution to conversion probability, ranking fourth by mean absolute SHAP value.

5.5.3 Incremental Feature Contribution Analysis

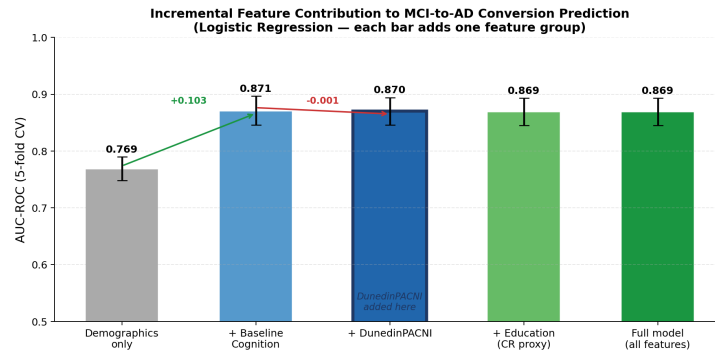


Figure 7: Incremental feature contribution (ablation) analysis showing changes in cross-validated AUC-ROC as predictor groups are added sequentially to the logistic regression model.

The ablation analysis revealed a pattern central to the theoretical interpretation of the full results. A demographic-only model achieved an AUC of 0.769. The largest single improvement was obtained by adding baseline cognition, increasing AUC by 0.103 to 0.871, demonstrating that contemporaneous cognitive performance provided by far the greatest discriminative gain within the feature set. Subsequent addition of DunedinPACNI resulted in a negligible reduction in performance (AUC = 0.870), and the addition of education produced a further minimal decrease, yielding a final seven-feature model AUC of 0.869. This pattern closely mirrors the cognition-adjusted logistic regression findings, suggesting that the predictive information captured by DunedinPACNI is largely subsumed by baseline cognitive status.

5.6 Summary

DunedinPACNI significantly predicted three-year conversion (OR = 1.25, $p = .022$), with effects remaining stable across all three primary logistic regression models. The primary models achieved AUC = 0.668–0.673. Education did not independently predict conversion ($p = .274$) and did not moderate the DunedinPACNI–conversion relationship (LRT $p = .441$). APOE $\epsilon 4$ was the strongest predictor (OR = 3.13, $p < .001$). In cognition-adjusted supplementary analyses, DunedinPACNI

was no longer significant ($p = .882$), indicating shared variance with baseline disease severity. The machine learning analysis achieved $AUC = 0.849\text{--}0.870$; the ablation analysis demonstrated that baseline cognition accounted for the overwhelming majority of the discriminative gain, whilst DunedinPACNI contributed negligible marginal AUC once cognition was included in the model.

6 Discussion

6.1 Principal Findings

DunedinPACNI significantly predicted three-year MCI-to-AD conversion across all three primary logistic regression models, with a per-standard-deviation odds ratio of 1.25 (95% CI: 1.03–1.51, $p = .022$) that was stable to inclusion of educational attainment, its interaction term, and established demographic and genetic covariates. APOE $\epsilon 4$ remained the dominant predictor at $OR = 3.13$ (95% CI: 2.18–4.54, $p < .001$) across all models. These findings answer the primary research question affirmatively: DunedinPACNI contributes independent prognostic information in MCI populations beyond the most important known genetic risk factor. The more theoretically consequential finding is what happens when baseline cognitive performance is introduced as a covariate: DunedinPACNI's contribution is attenuated to non-significance (SA1: $OR = 0.98$, $p = .882$), and this attenuation was independently confirmed by both the cognition-adjusted analyses and the machine learning ablation analysis, indicating that DunedinPACNI provides little incremental prognostic information once baseline cognitive performance is known.

6.2 DunedinPACNI: Biological Marker or Disease Severity Proxy?

The significant primary-model effect ($OR = 1.25$, 95% CI: 1.03–1.51, $p = .022$) is clinically meaningful and theoretically coherent. The stability of the PACNI effect across Models 1, 2, and 3 (OR range: 1.24–1.25, all $p \leq .025$) demonstrates that it captures conversion-relevant variance orthogonal to educational attainment. The cognition-adjusted findings (SA1: $OR = 0.98$, $p = .882$) require careful interpretation rather than dismissal. Two mechanisms could produce this pattern. Under

a mediation account, biological ageing pace accelerates neurodegeneration, which in turn manifests as early cognitive deterioration; adjusting for cognition therefore removes part of the pathway through which PACNI's effects on conversion are expressed. Under an overlap account, both DunedinPACNI and the cognition composite are imperfect indices of the same underlying disease severity construct. A third account, reverse causation, in which early subclinical cognitive decline drives the structural MRI changes DunedinPACNI captures, predicts identical attenuation and would imply that DunedinPACNI carries no independent biological signal at any stage. These accounts cannot be distinguished from cross-sectional data alone and require longitudinal measurement of DunedinPACNI trajectories alongside serial cognitive assessment.

6.3 Comparison with Previous DunedinPACNI Research

The present findings partially replicate and non-trivially refine the original Whitman et al. (2025) ADNI validation. Whitman et al. reported a hazard ratio of 1.44 (95% CI: 1.26–1.65) for MCI-to-dementia conversion using an ADNI MCI sample ($n = 701$) in a time-to-event Cox model. The present study's primary-model odds ratio of 1.25 (95% CI: 1.03–1.51) is meaningfully smaller; three design differences account for most of this attenuation. First, restricting the outcome to binary three-year conversion, and requiring minimum observed follow-up, preferentially excludes late converters and less-engaged participants, introducing a healthy volunteer bias that attenuates the conversion rate and therefore the effect estimate. Second, the binary logistic framework cannot account for censoring or temporal ordering of follow-up visits. Despite these effect size differences, the present study makes a distinct contribution: it establishes that DunedinPACNI's primary-model predictive contribution is substantially shared with baseline clinical severity, a finding not reported by Whitman et al., and that this attenuation is convergently confirmed by both parametric and algorithmic analyses of the same 614 participants.

6.4 The Null Effect of Education: Informative Failure

The absence of a significant independent association between education and conversion risk should not be interpreted as evidence that cognitive reserve is irrelevant

to AD progression. Accordingly, the hypothesised protective effect of educational attainment (H2) was not supported in the present cohort, although the estimated effect remained in the predicted direction (OR = 0.91). Likewise, the hypothesised moderation of the DunedinPACNI–conversion relationship by education (H3) was not supported (interaction OR = 1.07, $p = .441$). Four non-mutually-exclusive explanations must be considered. First, restriction of range in educational attainment reduces available statistical power: with a mean of 16.0 years and SD = 2.8, the sample is heavily concentrated in the upper educational range, and McClelland and Judd (1993) demonstrate that such truncation can reduce power to detect a moderation effect by 60–70%. Second, years of formal education is an inadequate operationalisation of a multidimensional construct that encompasses occupational complexity, leisure engagement, and social connectivity. Third, cognitive reserve may be substantially depleted by the time MCI is diagnosed, such that even high-reserve individuals show limited residual protective capacity. Fourth, the null finding may accurately reflect a genuine absence of education-mediated reserve effects within a three-year conversion window. These explanations may each contribute, and only adequately powered studies with greater educational diversity and composite reserve measurement could adjudicate among them.

6.5 APOE ϵ 4 and DunedinPACNI: Non-Redundant Biological Pathways

APOE ϵ 4 carrier status was the strongest predictor across all models (OR = 3.13, 95% CI: 2.18–4.54, $p < .001$), consistent with its established role as the primary genetic risk factor for late-onset AD (Corder et al., 1993; Liu et al., 2013). Its effect remained stable across all three primary models (OR range: 3.11–3.13), demonstrating independence from both DunedinPACNI and education within the present modelling framework. This statistical stability suggests that APOE ϵ 4 and DunedinPACNI contribute non-redundant predictive information for conversion risk. However, whether these predictors reflect genuinely distinct biological pathways cannot be determined from the present observational data and requires mechanistic investigation beyond the scope of this study.

6.6 Convergence of Statistical and Machine Learning Evidence

The ablation analysis demonstrated that adding cognition to demographic predictors increased AUC by approximately 0.103, whereas the incremental contribution of DunedinPACNI after cognition was already present was negligible (-0.001 AUC). This pattern closely mirrors the supplementary cognition-adjusted analyses, and the convergence from two independent analytic frameworks, one parametric, one algorithmic, substantially strengthens the conclusion that DunedinPACNI's primary-model association is largely shared with baseline disease severity. Across multiple feature importance analyses, DunedinPACNI consistently ranked third or fourth, supporting its characterisation as a statistically reliable predictor of conversion risk whilst the extent to which this reflects true multi-system biological ageing pace versus disease-related neuroimaging changes in an MCI sample remains to be established.

6.7 Robustness and Sensitivity of Findings

The primary finding appears robust by several criteria. The odds ratio was stable across three nested logistic regression models with different covariate sets (OR range 1.24–1.25), remained significant in the presence of the interaction term ($p = .025$ in M3), and its confidence interval did not overlap with 1.0 in any primary model. The convergence with machine learning feature importance analyses, where DunedinPACNI ranked fourth consistently across Gini, permutation, and SHAP analyses (Figures 5–6), provides multi-method corroboration. The stability of the APOE $\epsilon 4$ effect (OR range: 3.11–3.13 across all models) additionally confirms that multicollinearity is not artificially inflating individual estimates.

Several sensitivity-related uncertainties warrant discussion. The CSF biomarker subsample ($n = 339$) showed non-significant PACNI effects and a somewhat attenuated APOE $\epsilon 4$ effect (OR = 2.53 vs 3.13 in the full cohort), suggesting possible selection effects in the subset who underwent lumbar puncture. The severe multicollinearity between total tau and p-tau181 ($r = .984$, $VIF > 25$) renders individual CSF coefficients uninterpretable. Subgroup analyses stratifying by MCI subtype were not pre-specified and were deferred: with 211 Early MCI participants (25 converters, 11.8% conversion rate), the events-per-variable ratio for a four-predictor stratified model falls marginally below the accepted EPV threshold of 10, making

inference from such analyses unreliable in the present cohort.

6.8 Limitations

Several limitations constrain interpretation. First, the ADNI cohort is a highly selected, predominantly White, highly educated, health-motivated volunteer sample; this selection bias limits generalisability and specifically undermines the power to detect cognitive reserve effects. Second, DunedinPACNI was trained on a predominantly European-ancestry New Zealand cohort and validated in predominantly White UK and US samples; cross-ethnic validity requires independent validation before clinical deployment in ethnically diverse populations. Third, the median maximum observed follow-up month of 24 means that conversion status was determined from a single post-baseline visit for the majority of participants, creating non-trivial outcome misclassification risk for participants who converted between visits. Fourth, years of formal education is an inadequate reserve proxy in a high-attainment sample; composite reserve indices incorporating occupational complexity, leisure engagement, and social connectivity would better capture the construct. Fifth, the binary conversion outcome discards information about the rate and trajectory of decline; a time-to-event analysis would yield more statistical power and more clinically meaningful effect estimates. Sixth, the cross-sectional baseline design precludes causal inference; longitudinal measurement of DunedinPACNI trajectories is required to distinguish the mediation and overlap accounts of the cognition-adjustment attenuation. Seventh, amyloid or tau PET imaging data were unavailable for the majority of the cohort; adjusting for amyloid burden would provide a stronger test of DunedinPACNI's independence from Alzheimer's pathology specifically.

6.9 Clinical Implications and Future Directions

DunedinPACNI requires no additional scanning protocols, biofluid collection, or radiation exposure beyond established neuroimaging workflows. In the context of treatment eligibility assessment for anti-amyloid immunotherapies, where MRI is already mandatory for ARIA monitoring, DunedinPACNI computation adds no patient burden, representing a genuine practical advantage over blood-based biological ageing biomarkers and amyloid PET imaging. However, the cognition-

adjustment findings suggest that DunedinPACNI's incremental clinical value in established MCI populations, over and above standard cognitive assessment, is modest. This redirects its most promising application to pre-MCI populations where cognitive testing has not yet become abnormal but biological ageing pace may already be elevated, providing an early warning signal that motivates intensified monitoring or earlier preventive intervention.

Replication in larger, more educationally and ethnically diverse cohorts is the most immediate requirement. Testing the DunedinPACNI $\times$ Reserve interaction using composite reserve indices incorporating occupational complexity, leisure engagement, and social connectivity alongside education is likely to provide more sensitive tests of the buffering hypothesis. A formal mediation analysis using time-varying DunedinPACNI measurements, with cognitive performance as the proposed mediator, is needed to resolve the mediation versus overlap question. Future work should prioritise replication before these more ambitious designs.

7 Conclusions

This dissertation addressed a specific and previously untested question: whether neuroimaging-based biological ageing pace, indexed by DunedinPACNI, independently predicts three-year MCI-to-AD conversion and whether educational attainment moderates this relationship in ADNI. The primary finding is affirmative on the first question and inconclusive, not negative, on the second and third. Placing these results in their full context yields four distinct contributions.

First, DunedinPACNI significantly predicts three-year conversion from MCI to Alzheimer's disease (OR = 1.25, 95% CI: 1.03–1.51, $p = .022$), with the effect stable across three nested logistic regression models and independent of APOE $\epsilon 4$ status, age, sex, educational attainment, and their interaction. This result replicates the primary finding of Whitman et al. (2025) in a more restricted cohort and a different statistical framework, and provides convergent validation of DunedinPACNI's prognostic validity. The smaller effect size relative to the original hazard ratio of 1.44 is plausibly attributable to the binary versus time-to-event outcome, the restricted follow-up window, and the minimum visit requirement, though genuine effect modification by cohort characteristics cannot be excluded.

Second, the present study establishes an important and previously unreported

constraint on DunedinPACNI's clinical interpretation: its predictive contribution in MCI samples is substantially shared with baseline clinical severity. The attenuation from $OR = 1.25$ ($p = .022$) to $OR = 0.98$ ($p = .882$) following baseline cognition adjustment, independently confirmed by the machine learning ablation analysis (marginal AUC -0.001), constrains clinical deployment. In populations where comprehensive cognitive assessment is available — as it is in any MCI evaluation — DunedinPACNI provides negligible incremental prognostic information. Whether its utility extends to pre-MCI populations cannot be inferred from the present MCI data and requires prospective longitudinal studies beginning in midlife.

Third, the null findings for education as a main effect ($OR = 0.91$, $p = .274$) and as a moderator ($OR = 1.07$, $p = .441$) constitute informative evidence about the limits of ADNI as a reserve-testing cohort. The educational distribution ($M = 16.0$ years, $SD = 2.8$; effective range approximately 12–20 years) truncates precisely the portion of the distribution — low-to-moderate attainment — that drives reserve effects in population-representative studies. McClelland and Judd (1993) demonstrate this formally: restriction of range in a continuous predictor reduces power to detect moderation by approximately 60–70% in this scenario. Future tests of the DunedinPACNI $\times$ Reserve interaction must employ educationally diverse samples, composite reserve indices, and sample sizes specifically powered for interaction detection.

Fourth, the machine learning analysis demonstrates that strong discriminative performance in MCI conversion prediction (AUC 0.849–0.870 under nested cross-validation) is principally attributable to baseline cognitive performance rather than to biological ageing pace or algorithmic complexity. An interpretable logistic regression model matches the performance of ensemble methods on this feature set, suggesting that the computational overhead of XGBoost or Random Forest is not justified when a well-specified cognitive composite is available.

Looking forward, the most urgent single priority is a prospective pre-MCI longitudinal cohort study measuring DunedinPACNI serially before cognitive symptoms appear. This design alone would distinguish the mediation account, where PACNI predicts subsequent cognitive trajectory before any clinical impairment, from the overlap account, where PACNI and cognition remain tightly coupled regardless of disease stage. Resolving this question is prerequisite to understanding whether DunedinPACNI has genuine independent clinical value in early detection. The

path forward requires this prospective longitudinal evidence, combined with diverse cohorts and composite reserve operationalisation — a research agenda to which the present study provides a theoretically grounded and methodologically precise foundation.

8 References

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8.1 Appendix:Additional Visualisations

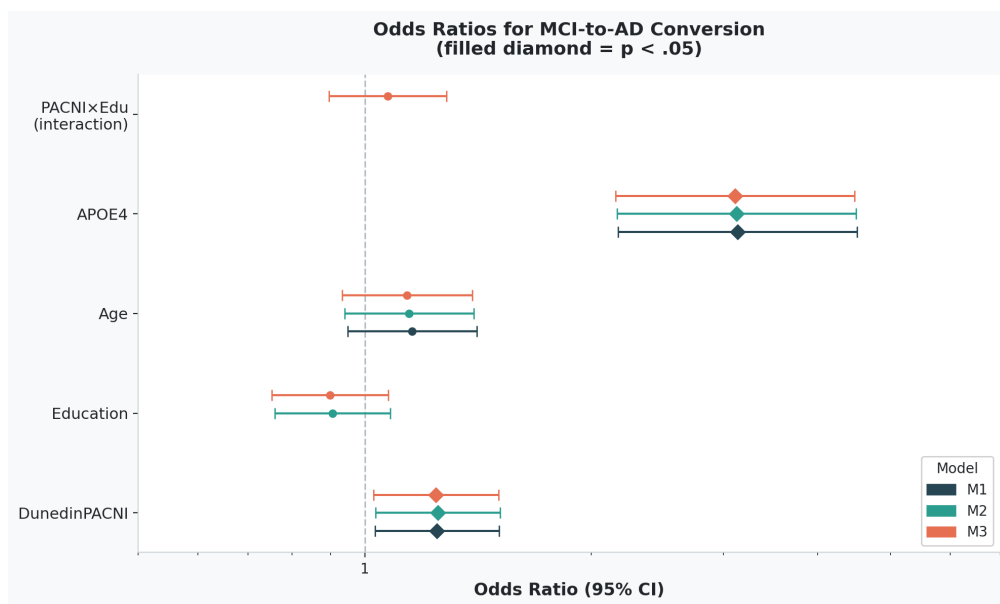


Figure 8: Points represent odds ratio estimates; horizontal bars represent 95% confidence intervals (Wald method). Dashed vertical line at $OR = 1.0$ represents no association. Log scale. *** $p < .001$.

Figure 8 presents odds ratio estimates and 95% confidence intervals for the three primary logistic regression models. DunedinPACNI estimates are stable across all three models and confidence intervals remain above 1.0, confirming robustness to the addition of education and the interaction term. APOE 4 is the dominant predictor ($OR = 3.1$) and remains stable alongside DunedinPACNI, confirming the two operate through non-redundant pathways. Notably, education (M2) and the

interaction term (M3) display wide confidence intervals spanning 1.0 — better interpreted as inconclusive under restriction of range than as evidence of no association.

8.2 Education Distribution by Conversion Status

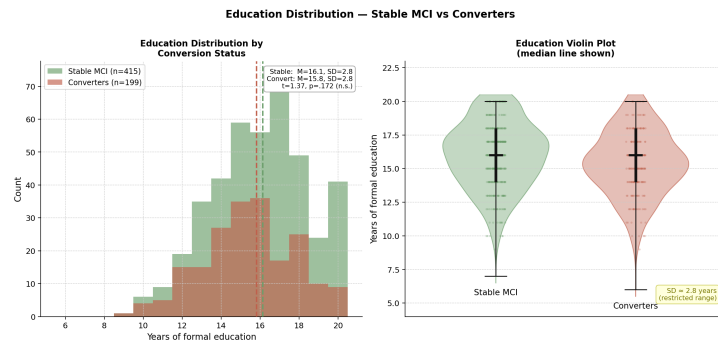


Figure 9: Left: overlapping histograms with group means indicated by dashed vertical lines. Right: violin plots with individual data points. The extremely similar distributions underscore the restricted educational range that limits the power to detect cognitive reserve effects in ADNI.

Figure 9 provides the single most important visual context for interpreting the education null finding. The two distributions, converters ($M = 15.8$, $SD = 2.8$) and stable MCI ($M = 16.1$, $SD = 2.8$), are so similar that the violin plots are nearly indistinguishable in shape, spread, and central tendency. The group difference of 0.3 years is not only statistically non-significant but substantively trivial. The critical methodological point visible in the histogram is the restriction of range. In a population-representative cohort, years of education would be expected to span roughly 0–20+ years with a standard deviation of 4–5 years, providing meaningful variation for statistical tests to detect a moderating effect. In this sample, the effective range is approximately 12–20 years, with very few participants below 12 years. This is a ceiling effect: almost the entire sample is in the upper portion of the education distribution, which is precisely where cognitive reserve theory predicts reserve will be least discriminating — the benefits of additional reserve accrue most substantially at lower levels of baseline reserve. McClelland and Judd (1993) provide the formal statistical basis for this concern. They demonstrate that

restriction of range in a continuous predictor can reduce the observable correlation between that predictor and an outcome by a factor proportional to the ratio of the restricted to unrestricted standard deviation squared. With a restricted SD of 2.8 versus a population SD of approximately 4–5 years, statistical power to detect a moderation effect of plausible magnitude is reduced by approximately 60–70%. The violin plot reinforces this: there is effectively no low-education tail in this sample — the individuals who, under reserve theory, would be most expected to show faster conversion in the presence of high biological ageing pace are largely absent from the ADNI cohort. This is not a failure of the theoretical framework; it is a sampling limitation that the present study is insufficiently powered to overcome.

8.3 Conversion Rate by APOE ϵ 4 Status and MCI Subtype

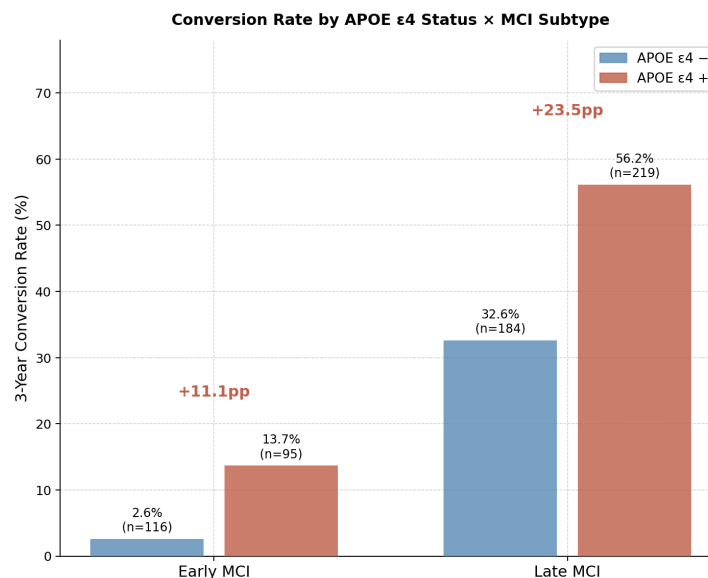


Figure 10: Values above bars show the conversion rate percentage and group n. Annotations show the percentage-point difference attributable to APOE ϵ 4 carriership within each MCI subtype.

Figure 10 reveals three substantively important patterns in the cross-tabulation of APOE ϵ 4 carrier status and MCI subtype. First, conversion rates are ordered as expected across all four cells — Early MCI APOE ϵ 4 negative shows the lowest

rate, and Late MCI APOE $\epsilon 4$ positive shows the highest — confirming that both factors are independently associated with progression risk and that their combination is particularly hazardous. Second, the absolute risk increment attributable to APOE $\epsilon 4$ carriership is larger in Late MCI than in Early MCI (approximately +24 vs +11 percentage points), suggesting that APOE $\epsilon 4$'s effect on conversion may be amplified when underlying neuropathological burden is already higher. This pattern is consistent with an accelerating rather than additive relationship between genetic risk and disease stage. Third, and directly relevant to the primary analysis: the pattern is approximately additive rather than strongly interactive. Late MCI without APOE $\epsilon 4$ has a substantially higher conversion rate than Early MCI with APOE $\epsilon 4$, indicating that MCI subtype captures disease burden that is partially independent of the genetic mechanism through which APOE $\epsilon 4$ operates. This supports the rationale for including both as independent predictors in the logistic regression models, and provides the visual foundation for understanding why DunedinPACNI adds predictive information above APOE $\epsilon 4$ alone: biological ageing pace is correlated with neither APOE $\epsilon 4$ status nor MCI subtype in a way that makes it redundant (VIFs all < 1.3), meaning it captures a third, partially independent dimension of conversion risk. The clinical implication is direct: a patient who is APOE $\epsilon 4$ negative and Early MCI carries only a 9% three-year conversion risk in this cohort. A patient who is APOE $\epsilon 4$ positive and Late MCI carries nearly a 50% risk. DunedinPACNI provides additional stratification within each of these cells — not replacing APOE $\epsilon 4$ or MCI subtype as risk factors, but adding information that these established markers do not capture.

8.4 DunedinPACNI vs Baseline Cognition by Conversion Status

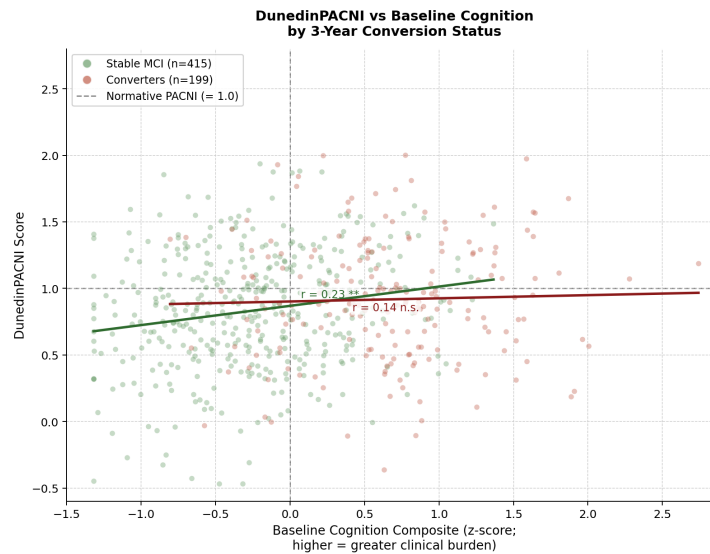


Figure 11: Scatter plot of DunedinPACNI score against baseline cognition composite (z-score, higher = greater clinical burden) by three-year conversion status. Lines show within-group linear regression fits with Pearson correlation coefficients. The dashed horizontal line at DunedinPACNI = 1.0 represents normative biological ageing pace. The partial overlap between these measures underpins the cognition-adjusted supplementary analysis findings.

Figure 11 addresses the central mechanistic question raised by the cognition-adjusted supplementary analyses: why does DunedinPACNI attenuate from $OR = 1.25$ ($p = .022$) to $OR = 0.98$ ($p = .882$) when baseline cognition is added? Two accounts are consistent with this pattern. Under a mediation account, biological ageing pace accelerates neurodegeneration, which manifests as early cognitive deterioration; adjusting for cognition therefore removes part of the pathway through which PACNI operates. The positive within-group correlations (converters: $r = 0.23$; stable MCI: $r = 0.14$) are consistent with this, as is the fact that the relationship holds in both groups rather than only in converters. Under an overlap account, both measures are imperfect proxies for the same underlying disease severity construct; adjusting for one removes their shared variance, leaving

neither independently predictive. The partial group overlap in the PACNI dimension supports this account. These two explanations cannot be distinguished from cross-sectional data alone and require longitudinal DunedinPACNI measurement before MCI onset. What the figure establishes is that the shared variance between PACNI and the cognition composite, not a statistical artefact, is the mechanism driving the cognition-adjustment attenuation.

8.5 Confusion Matrices, Learning Curves, and Calibration

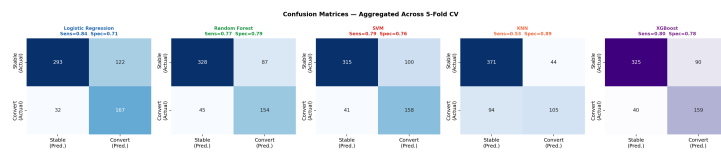


Figure 12: Confusion matrices aggregated across the five cross-validation folds. LR achieved the highest sensitivity (0.84); KNN achieved the highest specificity (0.89).

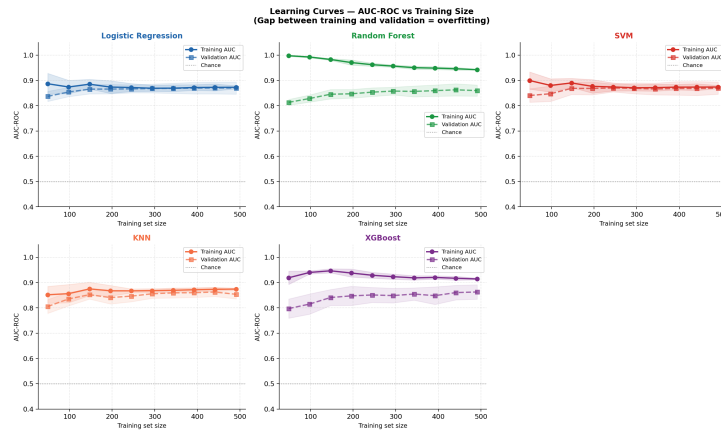


Figure 13: Learning set curves for all five classifiers: AUC-ROC as a function of training set size. Minimal train-validation gap for LR and SVM indicates good generalisation.

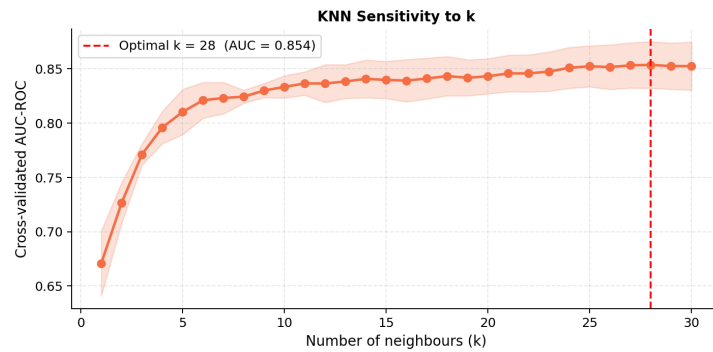


Figure 14: KNN sensitivity to neighbourhood size k . The k -sensitivity analysis identifies $k = 28$ as the value of peak cross-validated AUC (0.854). Note that the hyperparameter tuning inner fold selected $k = 21$; $k = 28$ represents the visual optimum of the post-hoc sensitivity sweep shown here.

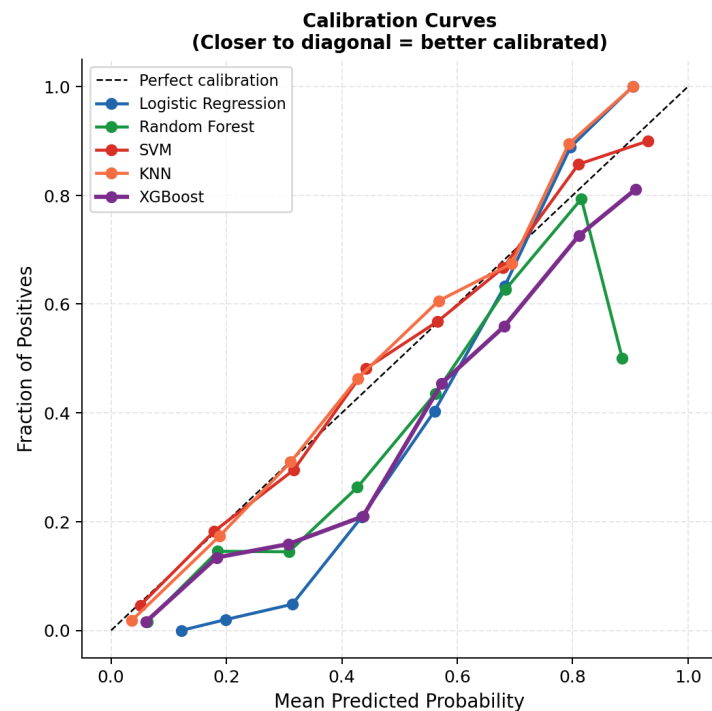


Figure 15: Calibration curves for all five ML classifiers. Logistic regression and SVM demonstrate the closest alignment with the calibration diagonal.

Confusion matrices aggregated across the five cross-validation folds (Figure 12) show that LR achieved the highest sensitivity (0.84) whilst KNN achieved the highest specificity (0.89), illustrating the precision-recall trade-off inherent in this imbalanced dataset. Learning curves (13) demonstrated that LR and SVM showed minimal train-validation gap, indicating good generalisation. XGBoost exhibited a persistent modest gap, suggesting marginal overfitting, though validation performance continued to improve with sample size. KNN sensitivity analysis (Figure 14) identified optimal $k = 28$ (AUC = 0.854). Calibration curves (Figure 15) showed that LR and SVM demonstrated the closest alignment with the calibration diagonal; XGBoost and RF exhibited moderate miscalibration at higher predicted probabilities.

8.6 Code Architecture

The analytical workflow followed a modular pipeline design comprising four sequential components: DunedinPACNI score generation, statistical modelling, cross-sectional biomarker validation, and machine learning evaluation. Each component was implemented as an independent, self-contained script with clearly defined inputs and outputs, supporting reproducibility and allowing any stage to be re-executed without rerunning the full pipeline.

The first stage, `run_DunedinPACNI.R`, computed participant-level DunedinPACNI biological ageing pace scores from pre-extracted MRI structural features. These scores constituted the primary predictor variable for all subsequent analyses and represented the entry point to the workflow.

The second stage, `analysis_pipeline.R`, merged DunedinPACNI scores with ADNI participant records, applied pre-specified inclusion criteria to construct the final 614-participant analytic cohort, derived the three-year MCI-to-AD conversion outcome, generated the baseline cognition composite, standardised all predictors, and executed the primary logistic regression models (M1–M3) alongside supplementary cognition-adjusted and CSF biomarker analyses, model diagnostics, ROC analyses, and calibration assessments. Its output, `final_cohort_clean.csv`, served as the input to the machine learning stage.

The third stage, `cn_mci_analysis.py`, provided cross-sectional validation of DunedinPACNI across cognitively normal (CN), early MCI (EMCI), late MCI (LMCI),

and Alzheimer’s disease (AD) groups using one-way ANOVA and Holm-corrected pairwise comparisons.

The fourth stage, `ml_comparison_pipeline.py`, compared five classifiers — Logistic Regression, Random Forest, SVM, KNN, and XGBoost — for predicting three-year conversion under a true nested five-fold cross-validation design, with `GridSearchCV` operating inside each outer fold to prevent optimism bias in reported performance estimates. The pipeline additionally produced feature importance analyses, SHAP explainability outputs, learning curves, calibration curves, and an ablation analysis quantifying the incremental contribution of each feature group.

A structured README documented software dependencies, execution order, input and output files, and key modelling decisions, enabling the complete workflow to be reproduced from DunedinPACNI generation through to machine learning evaluation.